

IOWA STATE UNIVERSITY
ELECTRICAL AND COMPUTER ENGINEERING

EE 2300

**LAB 1 REPORT – BASIC OPERATIONAL AMPLIFIER
CIRCUITS**

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Introduction

In this lab, we started working with different variations of op amp circuits. We started with a non-inverting and a inverting op amp circuit, where we calculated the expected gain of each circuit and compared it with our measured gain. For the non-inverting amp, we also observed the transfer characteristics on the oscilloscope. For the inverting amp, we calculated the gain with two different values for the frequency resistor. After that, we built a circuit with and without a unity-gain buffer set up and observed the difference between each setup. Then we built a integrating amplifier circuit and observed the input and output traces on the oscilloscope. We used three types of inputs for this which include Square Wave input, Ramp input and Sinusoidal input. After that, we built a differential op amp circuit, after which we measured the differential mode gain for each source while the other was grounded, then measured the common mode gain too. Then we compared them with our calculated values. Then we built a summing amplifier to create a digital-to-analog converter (DAC). We first had to figure out the resistor ratios and values, then choose specific values for each resistor to built the circuit. In the end, we built an instrumentation amp circuit to observe the common mode and differential mode gain, which we also observed traces for on the oscilloscope.

Experiment 1. Non-Inverting Amplifier

The table below lists the measurement results.

Parameter	Value
Calculated Gain (V/V)	16
Measured v_i (V)	0.5
Measured v_o (V)	7.97
Measured Gain (V/V)	15.94

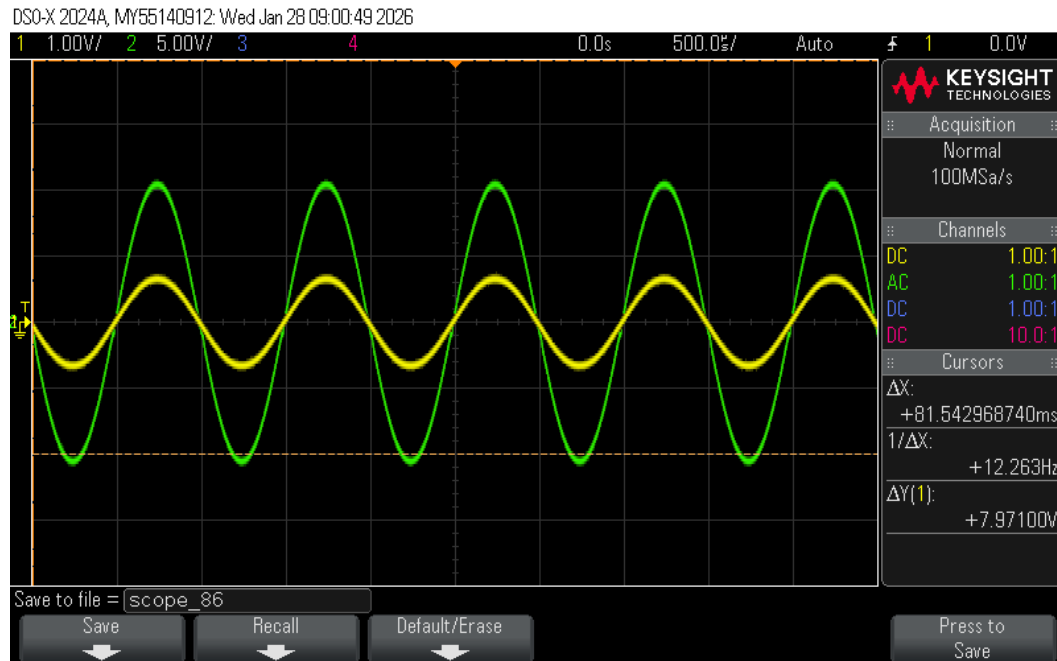


Figure 1-1. Oscilloscope trace of v_i (yellow) and v_o (green).

Experiment 2. Observing the Transfer Characteristic

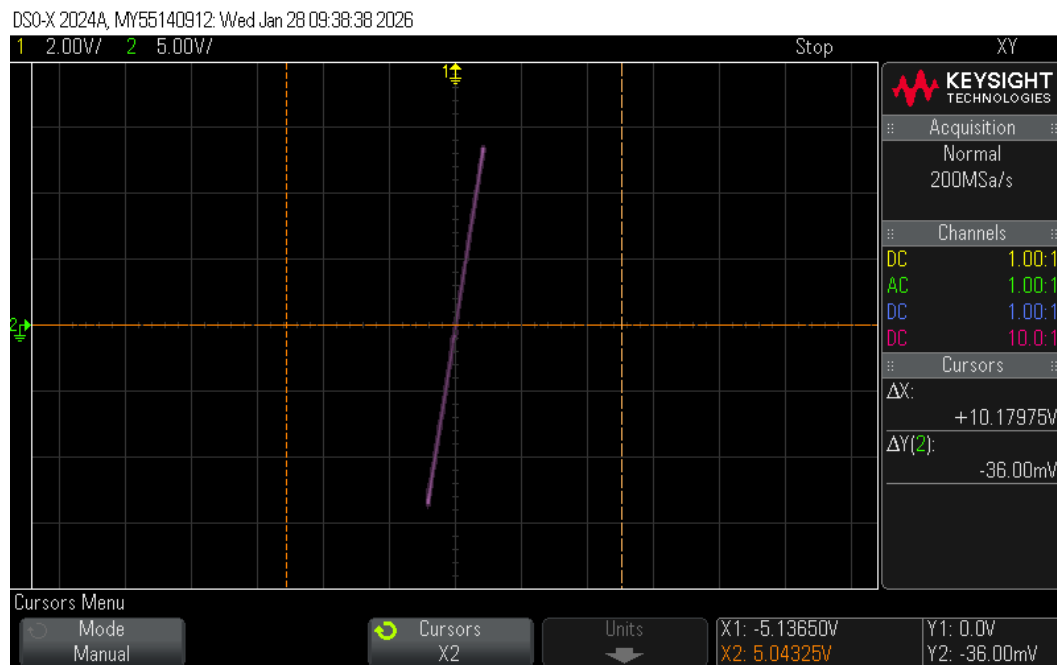


Figure 2-1. X-Y Plot of the input and output with no clipping.

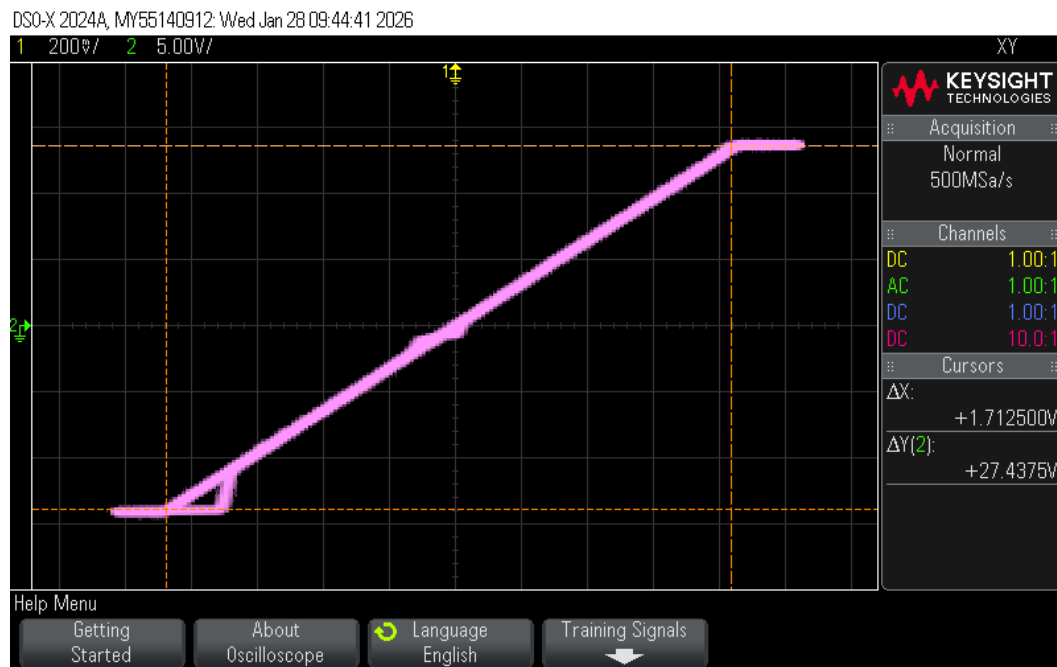


Figure 2-2. X-Y Plot of the input and output with clipping.

In figure 2-1, the op amp is operating in its linear region. The y-axis represents the output voltage (v_o) and the x-axis represents the input voltage (v_i). The slope v_o / v_i represents the gain of the op amp. In figure 2-1, v_o is under the source voltage of the op amp which is ± 15 V. So, we don't see any clipping.

As for figure 2-2, we see the line flattens out at the top and bottom. This shows that the output is exceeding the source voltage and is being clipped. But the gain/slope is still close to 16. But the input voltage keeps increasing. This happens because even if the output voltage is 16 times the input voltage, it can never exceed the supply voltage of the op amp.

Experiment 3. Inverting Amplifier

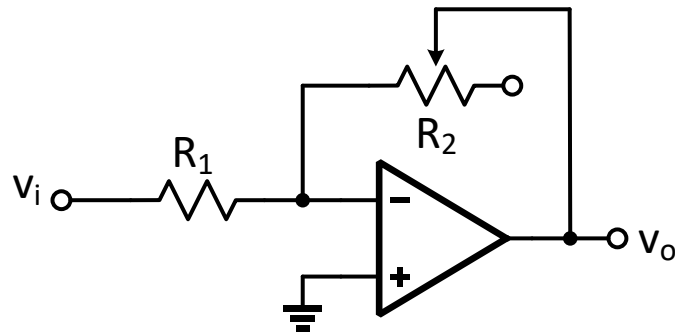


Figure 3-1. Schematic of the inverting amplifier used for this experiment. The amplifier uses an LMC660 op amp with ± 8 V supplies and $R_1 = 10\text{ k}\Omega$, $R_2 = 100\text{ k}\Omega$ potentiometer.

The table below lists all of the measurement values. All work for calculating the expected gain is attached to the end of this report.

Parameter	Value ($R_2 = 50\text{ k}\Omega$)	Value (Max. Gain)
Measured R_2 (k Ω)	49.288	59.7
Calculated Gain (V/V)	-22.72	-27.14
Measured v_i (V)	0.195	0.195
Measured v_o (V)	4.40	5.37
Measured Gain (V/V)	22.56	27.54

The gains we calculated are negative but close to the values we measured, which are positive. This indicates a phase shift of 180 degrees. The values we measured came out positive because the multimeter doesn't calculate the sign, only the magnitude.

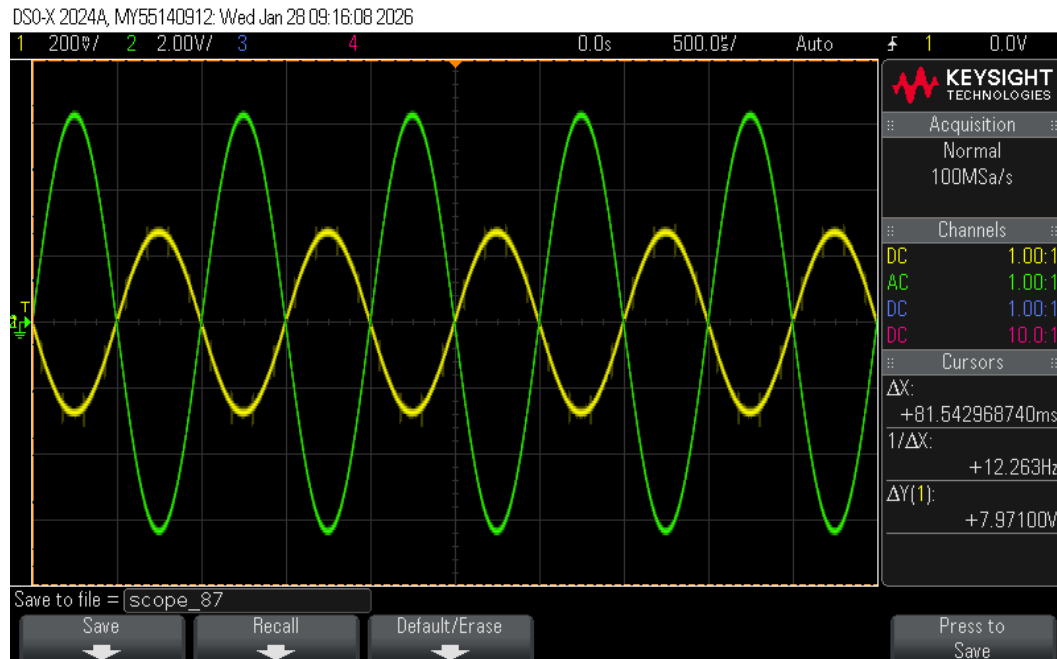


Figure 3-2. Oscilloscope trace of v_i (yellow) and v_o (green) with $R_2 = 50 \text{ k}\Omega$.

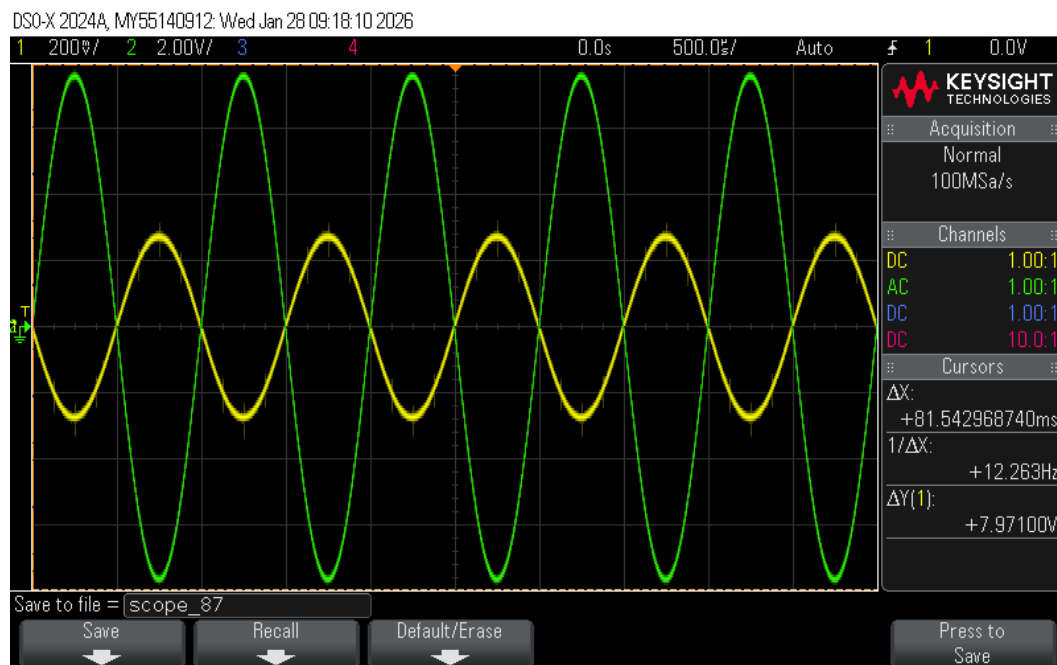


Figure 3-3. Oscilloscope trace of v_i (yellow) and v_o (green) with R_2 set for maximum gain.

If we were to put the potentiometer at the input instead of the frequency resistor, then that potentiometer would control the input voltage. It would either increase or decrease the input voltage which would in turn also impact on the output voltage. A higher potentiometer value would lower the overall gain, whereas a lower potentiometer value would increase the overall gain.

Experiment 4. Unity-Gain Buffer

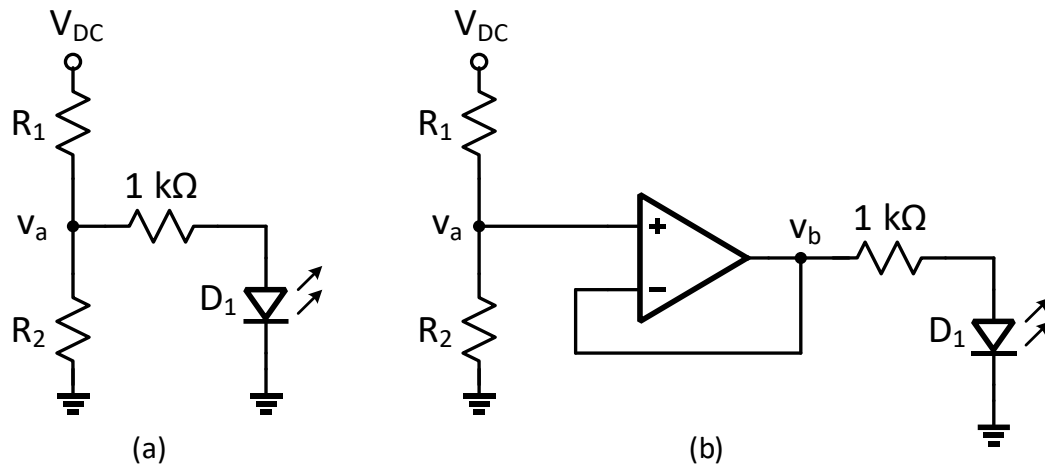


Figure 4-1. Schematic of (a) voltage divider loaded by a diode and (b) same voltage divider but with a unity-gain buffer inserted between the divider and the diode.

The table below lists the measurement results.

Parameter	Resistor Pair (R_1/R_2)		
	2.2/6.8 k Ω	4.7/4.7 k Ω	6.8/2.2 k Ω
Calculated V_A with no diode (V)	11.33	7.5	3.67
Calculated V_A of Circuit (a) (V)	5.66	4.09	2.76
Measured V_A of Circuit (a) (V)	5.47	3.59	2.54
Measured V_A of Circuit (b) (V)	11.3	7.46	3.65
Measured V_B of Circuit (b) (V)	11.3	7.46	3.65

Without the unity gain buffer, the led and resistor branch draws current, thus reducing the V_A . But with the unity gain buffer, the load on V_A is reduced as the op amp takes significantly reduced current, thus the V_A remains almost the same as the values we calculated without the diode.

Experiment 5. Integrating Amplifier

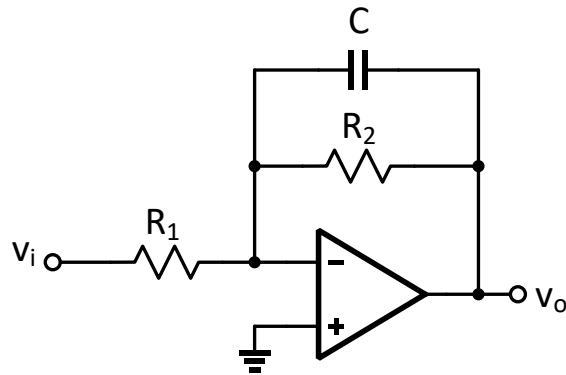


Figure. 5-1. Schematic of the integrating amplifier used for this experiment. The amplifier uses an LMC660 op amp with ± 8 V supplies and $R_1 = 10$ k Ω , $R_2 = 470$ k Ω , and $C = 0.1$ μ F.

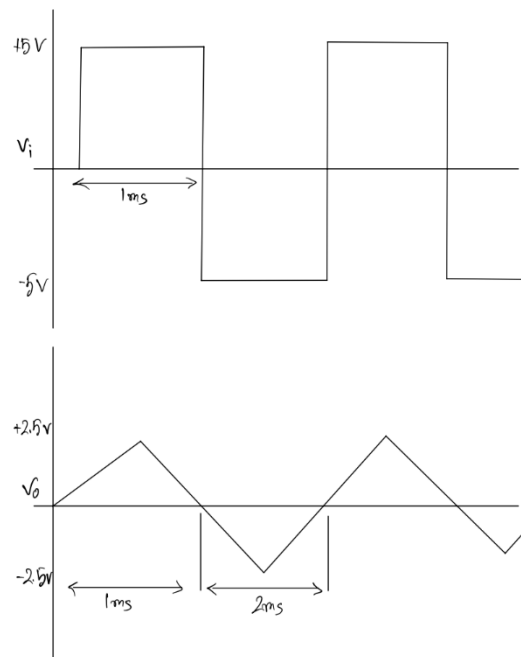


Figure 5-2. Sketch of the expected integrator output for a square-wave input.

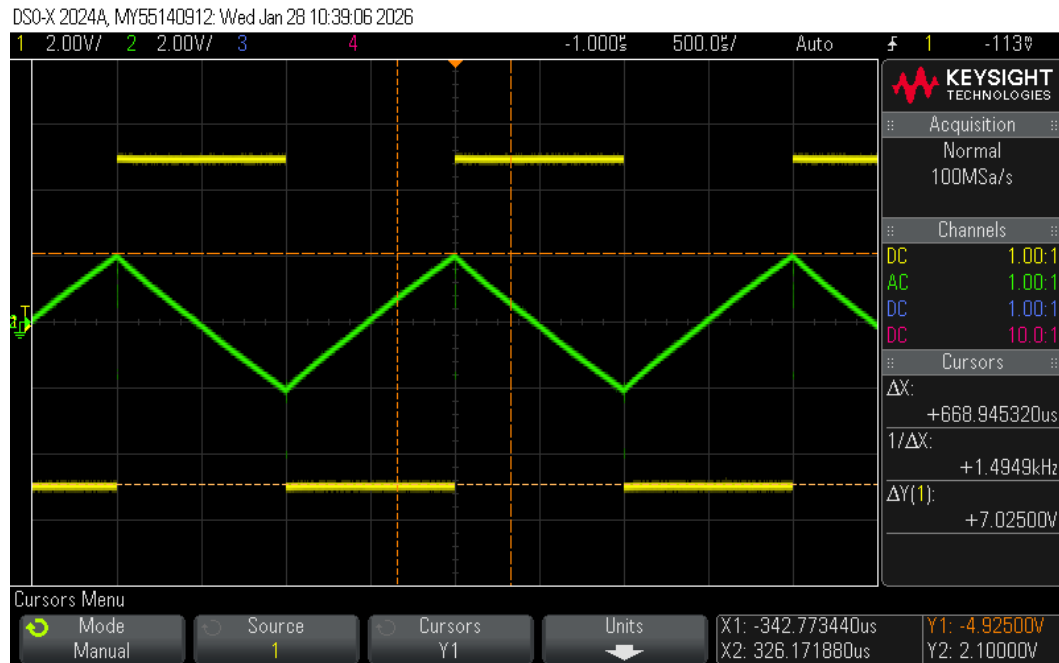


Figure 5-3. Oscilloscope trace of v_i (yellow) and v_o (green) using a square-wave as the input.

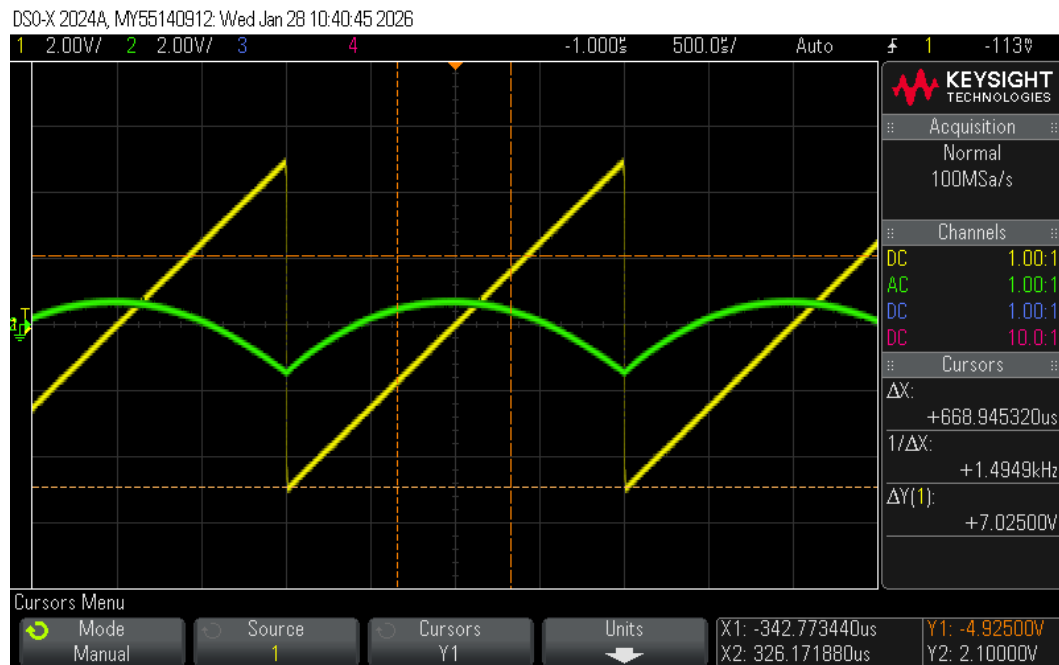


Figure 5-4. Oscilloscope trace of v_i (yellow) and v_o (green) using a ramp as the input.

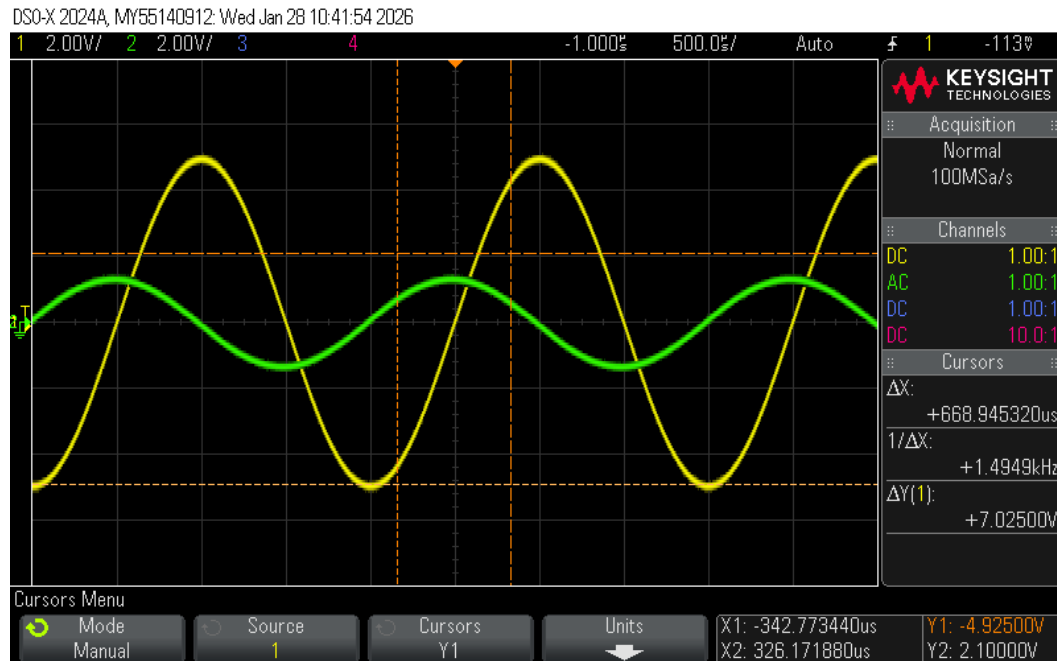


Figure 5-5. Oscilloscope trace of v_i (yellow) and v_o (green) using a sine-wave as the input.

The output signals are the integrals of the input signals.

In 5.3, the input signal is a square wave going from +5V to -5V. This constant shift between positive or negative voltage produces an integral which looks like a triangle because of the output voltage constantly and linearly increasing and decreasing.

In 5.4, the input signal is a ramp, which produces a straight line that is constantly changing. The integral of this linear line is a quadratic curve which can be seen in the output voltage.

In 5.5, the input signal is a sine wave, as such the output signal is also a sine wave but with a phase shift. The integral of a sine wave is another sine wave with a noticeable phase shift and a different amplitude.

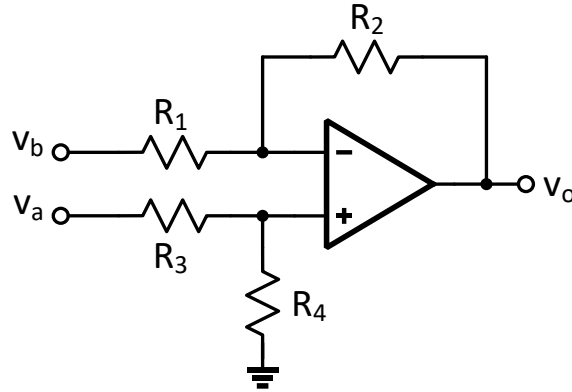
Experiment 6. Simple Difference Amplifier

Figure 6-1. Schematic of the difference amplifier used for this experiment. The amplifier uses an LMC660 op amp with ± 8 V supplies and $R_1 = 1\text{ k}\Omega$, $R_2 = 10\text{ k}\Omega$, $R_3 = 2.2\text{ k}\Omega$, and $R_4 = 22\text{ k}\Omega$.

The table below lists the measured resistor values.

Measured Resistance ($\text{k}\Omega$)	
R_1	0.985
R_2	9.79
R_3	2.168
R_4	21.7

The table below lists the measurement results.

Parameter	Value (v_b grounded)	Value (v_a grounded)	Value (with $v_a = v_b$)
Calculated Gain	9.945	9.939	0.006
Measured Gain	9.95	9.95	6.95×10^{-3}

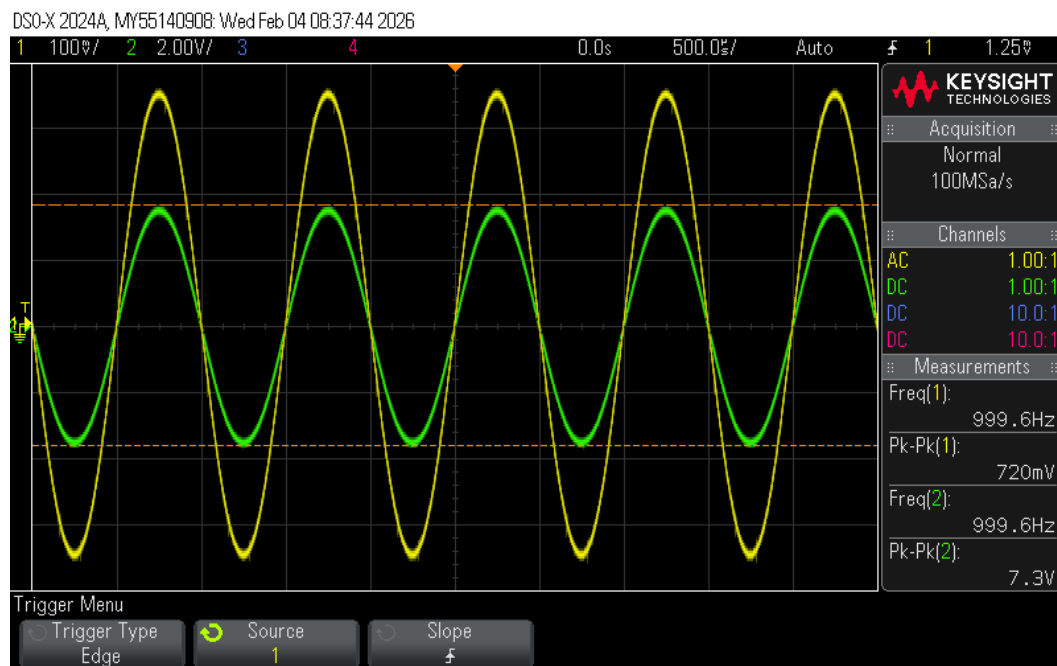


Figure 6-2. Oscilloscope trace of v_a (yellow) and v_o (green) with v_b grounded.

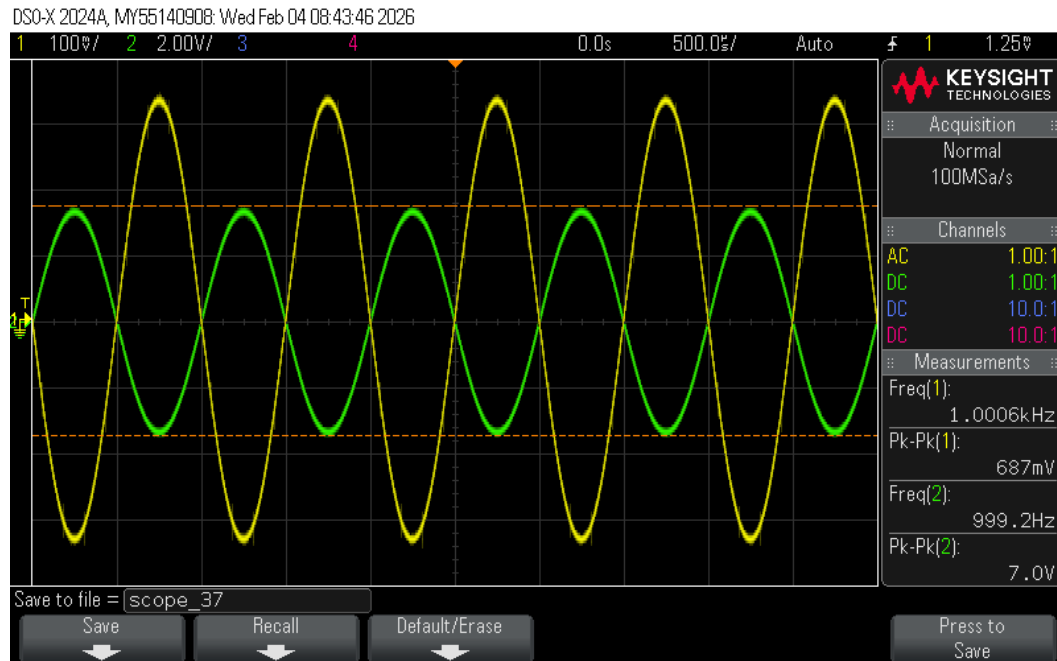


Figure 6-3. Oscilloscope trace of v_b (yellow) and v_o (green) with v_a grounded.

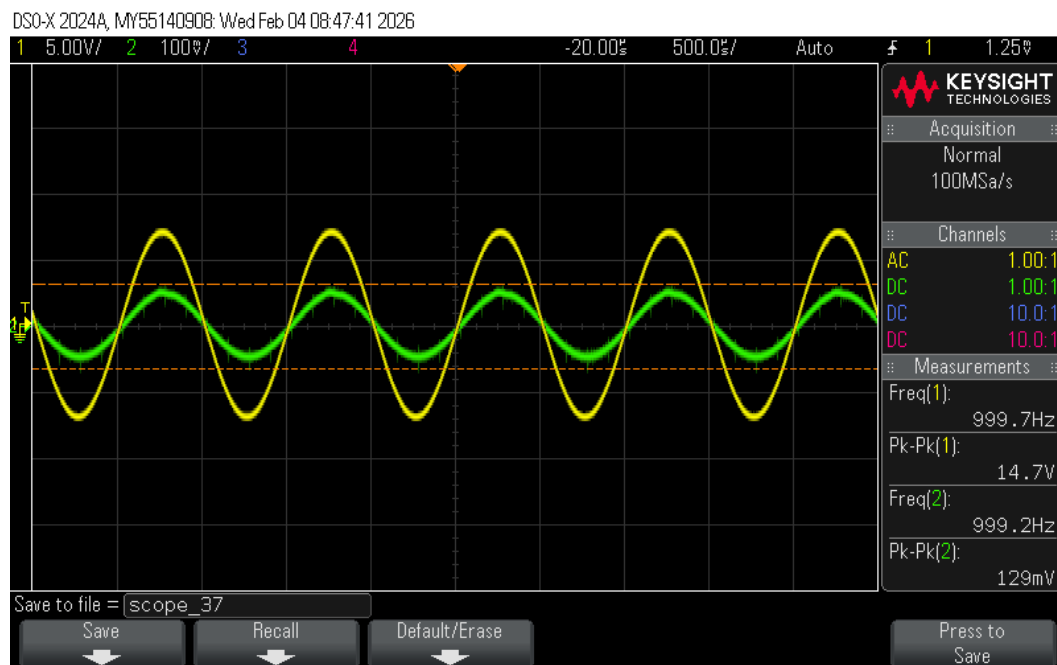


Figure 6-4. Oscilloscope trace of v_i (yellow) and v_o (green) with $v_a = v_b$.

The output is not zero when the outputs are connected because the circuit components are not exactly ideal. For example, in an ideal differential op amp circuit the ratios R_2/R_1 and R_4/R_3 are supposed to be equal. But in this real-life circuit, that is not the case.

Also, the op amp is not ideal either, as it has a small offset voltage. For these non-ideal properties the output voltage is not zero.

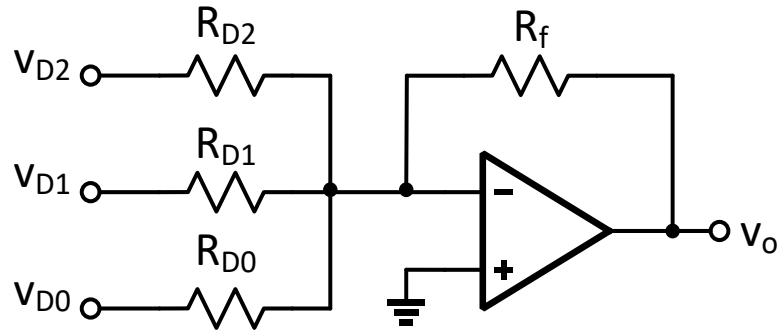
Experiment 7. Summing Amplifier

Figure 7-1. Schematic of summing amplifier used for this experiment.

The required resistor ratios are given below:

$$\begin{aligned}\frac{R_f}{R_{D2}} &= 4 \\ \frac{R_f}{R_{D1}} &= 2 \\ \frac{R_f}{R_{D0}} &= 1\end{aligned}$$

To obtain these ratios, the following resistor values were chosen:

Measured Resistance (k Ω)	
R_f	3.91
R_{D2}	0.984
R_{D1}	1.965
R_{D0}	3.9

The table below lists the measured output voltage for all 8 input combinations:

v_{D2} (V)	v_{D1} (V)	v_{D0} (V)	v_o (V)
0	0	0	0
0	0	1	-1.02
0	1	0	-2.01
0	1	1	-3.05
1	0	0	-4.05
1	0	1	-5.08
1	1	0	-6.09
1	1	1	-7.11

To design this circuit, we first use three test values (101=-5, 110=-6, 011=-3) to figure out the resistor ratios by plugging the values into the output voltage equation for a summing amplifier. After we achieved the ratios ($R_f/R_{D2} = 4$, $R_f/R_{D1} = 2$, $R_f/R_{D0} = 1$), we set $R_f = 4 \text{ k}\Omega$ and figured out the other values ($R_{D2} = 1 \text{ k}\Omega$, $R_{D1} = 2 \text{ k}\Omega$, $R_{D0} = 4 \text{ k}\Omega$).

To achieve those values, we used a set of $1 \text{ k}\Omega$ resistors in series for each one. One challenge that we faced was figuring out the combinations at first to get out desired resistor value ratio. This had an easy solution, where we just used a lot of $1 \text{ k}\Omega$ resistors.

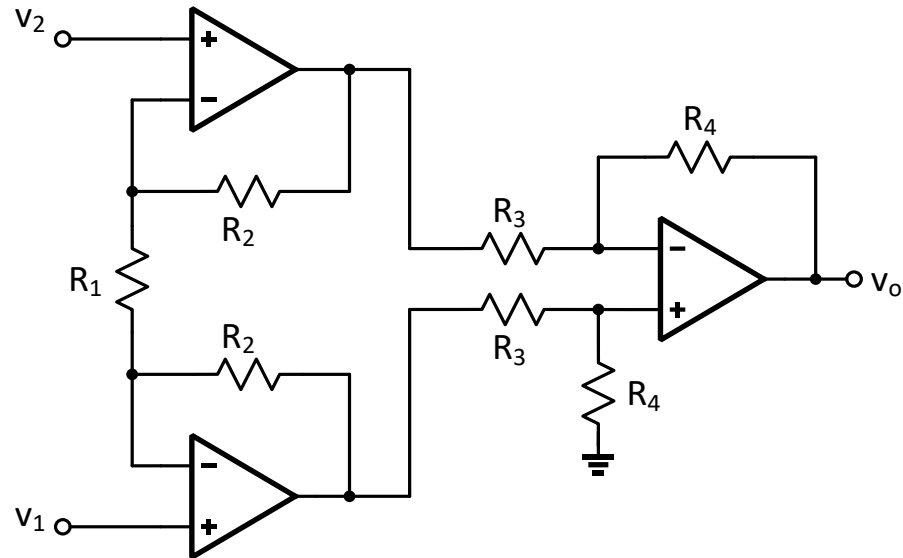
Experiment 8. Instrumentation Amplifier

Figure 8-1. Schematic of the instrumentation amplifier used for this experiment. The amplifier uses an LMC660 op amp with ± 8 V supplies and $R_1 = 10\text{ k}\Omega$, $R_2 = 10\text{ k}\Omega$, $R_3 = 1\text{ k}\Omega$, and $R_4 = 10\text{ k}\Omega$.

The table below lists the measurement results:

Parameter	Value (v_b grounded)
Calculated Differential Gain (V/V)	30
Measured Differential Gain (V/V)	29.75
Measured Common-Mode Gain (V/V)	0

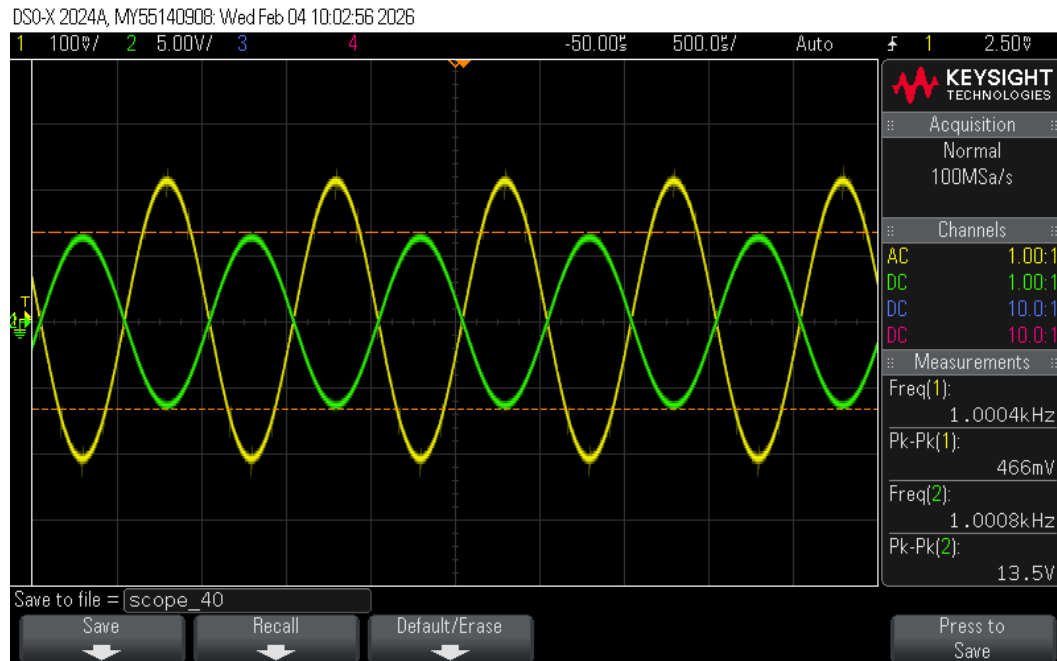


Figure 8-2. Oscilloscope trace of v_i (yellow) and v_o (green) showing differential gain (with v_2 grounded).

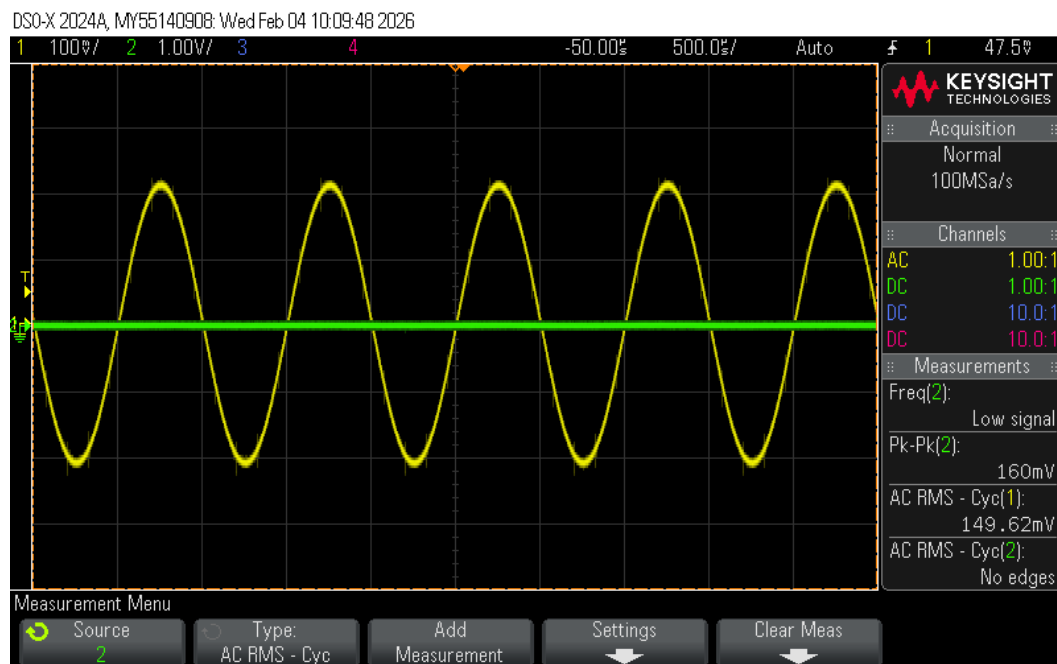


Figure 8-3. Oscilloscope trace of v_i (yellow) and v_o (green) showing common-mode gain (with $v_1=v_2$).

Conclusion

In this lab, we went over most things related to circuits we went through in 201. We built different op amp circuits, calculated different elements and measured them to see if they match our calculations. We also observed the input and output voltages on the oscilloscope and identified different properties of them.

The biggest problem in this lab that I faced was getting reacquainted with some of features of the lab equipment and how to understand them. Alongside that, we faced problems figuring out the resistor values that we would need to build the DAC in experiment 7. But we managed to figure out an easy solution to that by using the $1\text{ k}\Omega$ resistors only. Aside from these, the lab went pretty smoothly and I was able to go over what I learnt in 201 again.

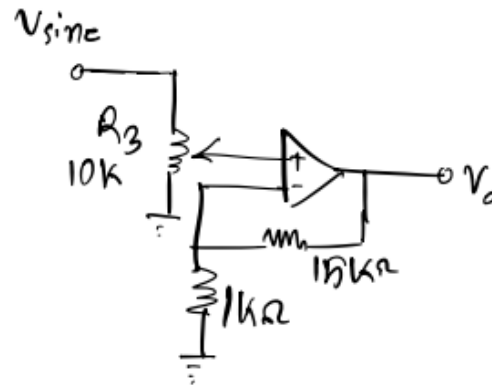
Calculations:Experiment 1:

For the non-inverting
amp,

$$G = \frac{V_o}{V_i} = \left(1 + \frac{R_f}{R_s}\right)$$

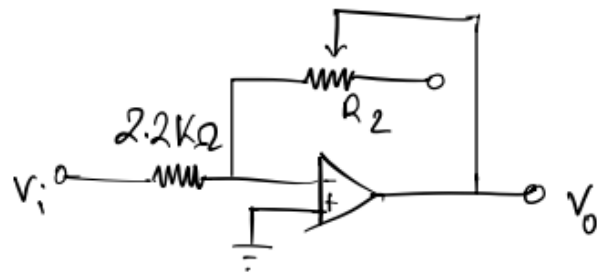
$$= (1 + 15)$$

$$= 16 \text{ V/V}$$

Experiment 3:

For,

$$R_2 = 50k\Omega$$



For inverting op amp,

$$\text{Gain, } \frac{V_o}{V_i} = -\frac{R_f}{R_s} = -\frac{50k}{2.2k} = -22.72 \text{ V/V}$$

For, $R_2 = 59.7k\Omega$ (Potentiometer value where output is at its maximum value without dipping)

$$\text{Gain, } \frac{V_o}{V_i} = -\frac{R_f}{R_s} = -\frac{59.7k}{2.2k} = -27.14 \text{ V/V}$$

Experiment 4:

a) Without the diode,
the $1k\Omega$ path is open,
so no current flows through
there

$$V_a = 15 \cdot \frac{R_2}{R_1 + R_2}$$

For, $R_1 = 2.2k\Omega$ & $R_2 = 6.8k\Omega$,

$$V_a = 15 \cdot \frac{6800}{2200 + 6800}$$

$$= 11.33 \text{ V}$$

For, $R_1 = 4.7k\Omega$ & $R_2 = 4.7k\Omega$,

$$V_a = 7.5 \text{ V}$$

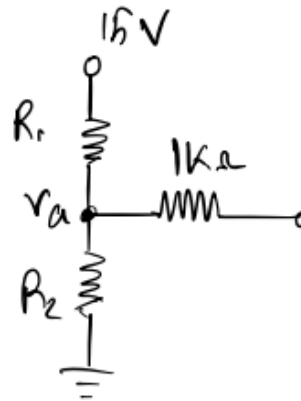
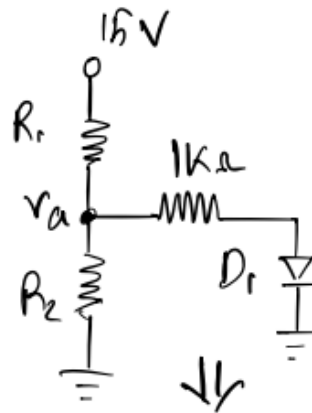
For, $R_1 = 6.8k\Omega$ & $R_2 = 2.2k\Omega$,

$$V_a = 3.67 \text{ V}$$

b) With the diode (LED) $\Rightarrow$

Using KCL,

$$-\frac{15 - V_a}{R_1} + \frac{V_a - 0}{R_2} + \frac{V_a - 2}{1000} = 0$$



$$\Rightarrow \frac{15 - v_a}{R_1} = \frac{v_a}{R_2} + \frac{v_b}{1000} - \frac{2}{1000}$$

$$\Rightarrow 15 - v_a = v_a \left(\frac{R_1}{R_2} + \frac{R_1}{1000} \right) - \frac{2R_1}{1000}$$

$$\Rightarrow 15 + \frac{2R_1}{1000} = \left(\frac{R_1}{R_2} + \frac{R_1}{1000} + 1 \right) v_a$$

$$\Rightarrow v_a = \frac{15 + \frac{2R_1}{1000}}{\frac{R_1}{R_2} + \frac{R_1}{1000} + 1}$$

For, $R_1 = 2.2\text{k}\Omega$ & $R_2 = 6.8\text{k}\Omega$,

$$v_a = \frac{15 + \frac{2200 \times 2}{1000}}{\left(\frac{2200}{6800} + \frac{2200}{1000} + 1 \right)} = 5.66\text{V}$$

For, $R_1 = 4.7\text{k}\Omega$ & $R_2 = 4.7\text{k}\Omega$,

$$v_a = 4.09\text{V}$$

For, $R_1 = 6.8\text{k}\Omega$ & $R_2 = 2.2\text{k}\Omega$,

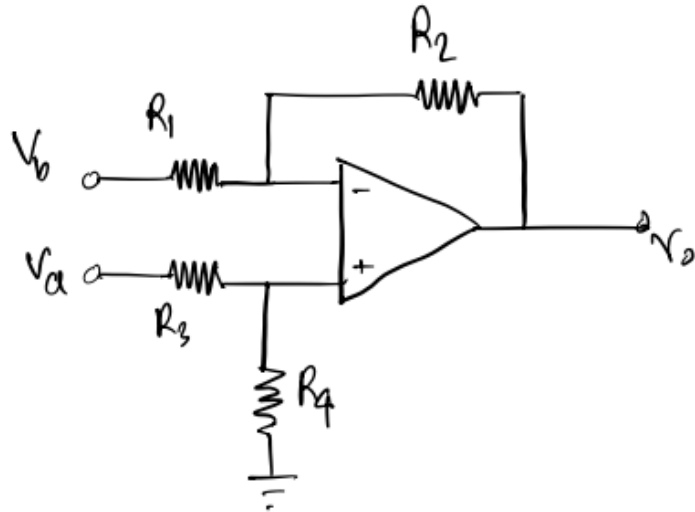
$$v_a = 2.76\text{V}$$

Experiment 6:

In the circuit,

$$\frac{R_2}{R_1} = 9.939$$

$$\frac{R_4}{R_3} = 10.009$$



$$V_{DM} = V_a - V_b$$

$$V_{cm} = \frac{V_a - V_b}{2}$$

when $V_b = 0$,

$$V_p = V_a \cdot \frac{R_4}{R_3 + R_4} \quad (\text{voltage divider})$$

$$G = \frac{V_o}{V_a} = \left(1 + \frac{R_2}{R_1}\right) \left(\frac{R_4}{R_3 + R_4}\right)$$

$$= 9.945$$

when $V_a = 0$,

$$V_o = -\frac{R_2}{R_1} V_b$$

$$G = \frac{V_o}{V_b} = -\frac{R_2}{R_1} = -9.939$$

$$|G| = 9.939$$

$$\text{Common mode gain } (V_a = V_b), G_{cm} = \left(1 + \frac{R_2}{R_1}\right) \left(\frac{R_4}{R_3 + R_4}\right) - \left(-\frac{R_2}{R_1}\right)$$

$$= 0.006$$

Experiment 4:

$$\frac{V_a}{R_a} + \frac{V_b}{R_b} + \frac{V_c}{R_c} = \frac{0 - V_o}{R_f}$$

$$V_o = - \left(V_a \cdot \frac{R_f}{R_a} + V_b \cdot \frac{R_f}{R_b} + V_c \cdot \frac{R_f}{R_c} \right)$$

To get -5 from input 101,
 $V_a = 1, V_b = 0, V_c = 1, V_o = 5$

we get,

$$\frac{R_f}{R_{D2}} + \frac{R_f}{R_{D0}} = 5 \quad \text{--- (i)}$$

Similarly, for 110,

$$\frac{R_f}{R_{D2}} + \frac{R_f}{R_{D1}} = 6 \quad \text{--- (ii)}$$

for 011,

$$\frac{R_f}{R_{D1}} + \frac{R_f}{R_{D0}} = 3 \quad \text{--- (iii)}$$

from (i), (ii) & (iii) $\Rightarrow$

$$\frac{R_f}{R_{D2}} = 4 \quad \frac{R_f}{R_{D0}} = 1$$

$$\frac{R_f}{R_{D1}} = 2$$

If we take, $R_f = 4k\Omega$
 then we need, $R_{D2} = 1k\Omega$, $R_{D1} = 2k\Omega$, $R_{D0} = 4k\Omega$